

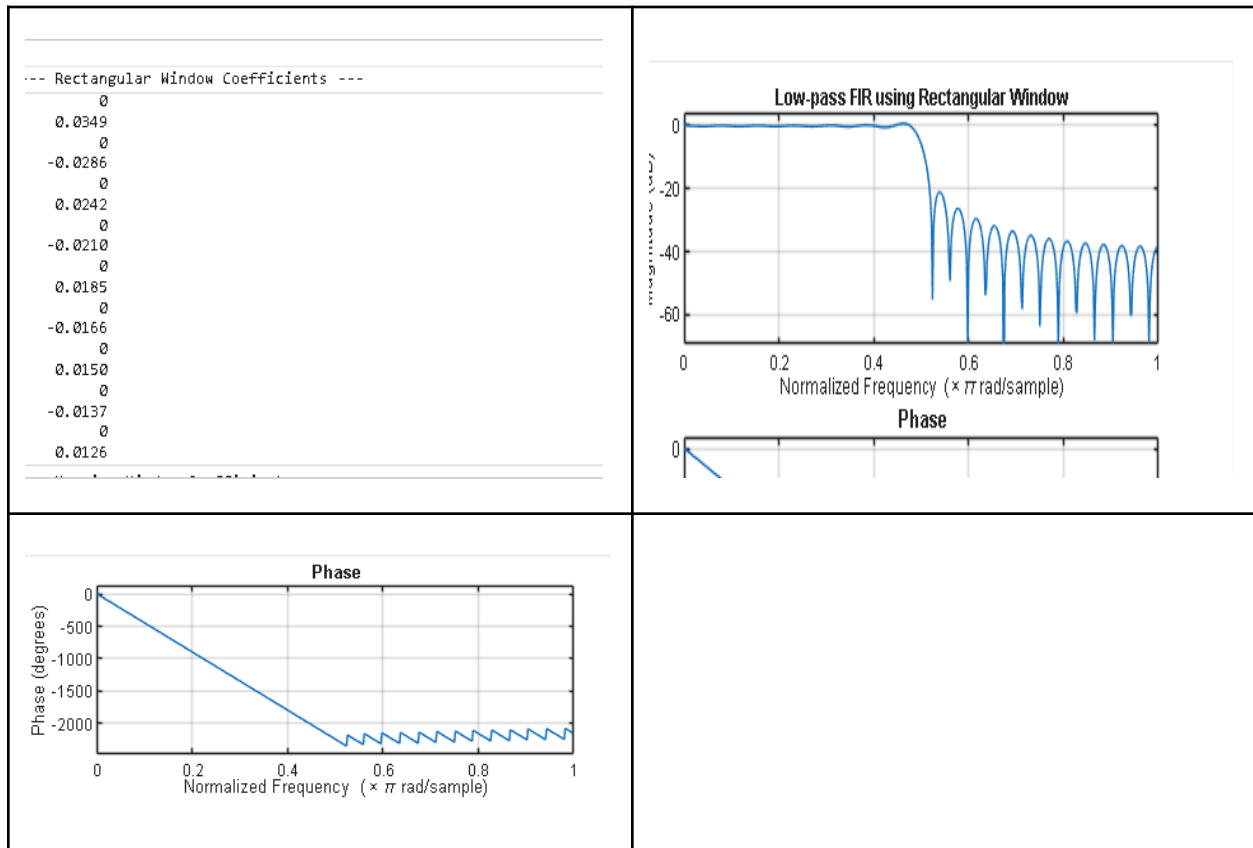
Expt 4: Design of Low Pass Filter using Windowing Techniques

4.1. Design and Analysis of an FIR Low-Pass Filter Using the Rectangular Window Technique in MATLAB

Program

```
clc;
clear;
close all;
%% --- Input Parameters ---
fp = 1000; % Passband cutoff (Hz)
fs = 4000; % Sampling frequency (Hz)
N = 50; % Filter order (even number preferred)
wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)
%% --- 1. FIR Low Pass using Rectangular Window ---
b_rect = fir1(N, wc, 'low', rectwin(N+1));
disp('--- Rectangular Window Coefficients ---');
disp(b_rect);
%% --- 2. FIR Low Pass using Hamming Window ---
b_hamm = fir1(N, wc, 'low', hamming(N+1));
disp('--- Hamming Window Coefficients ---');
disp(b_hamm);
%% --- 3. FIR Low Pass using Hanning Window ---
b_hann = fir1(N, wc, 'low', hanning(N+1));
disp('--- Hanning Window Coefficients ---');
disp(b_hann);
%% --- Plot Frequency Responses ---
figure;
freqz(b_rect,1);
title('Low-pass FIR using Rectangular Window');
figure;
freqz(b_hamm,1);
title('Low-pass FIR using Hamming Window');
figure;
freqz(b_hann,1);
title('Low-pass FIR using Hanning Window');
```

Output :



Expt 4 Design of Low Pass Filter using Windowing Techniques

4.2. Design and Analysis of an FIR Low-Pass Filter Using the Hamming Window Technique in MATLAB

Program :

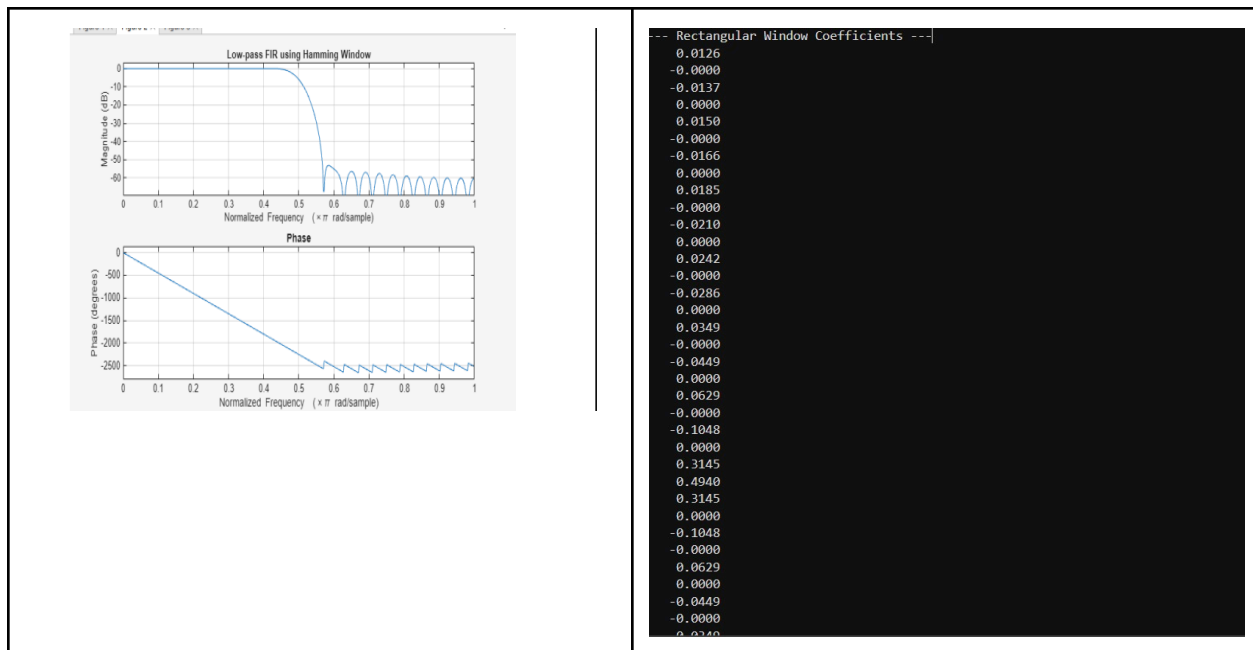
```
clc;
clear;
close all;
%% --- Input Parameters ---
fp = 1000; % Passband cutoff (Hz)
fs = 4000; % Sampling frequency (Hz)
N = 50; % Filter order (even number preferred)
wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)
```

```

%% --- FIR Low Pass using Hamming Window ---
b_hamm = fir1(N, wc, 'low', hamming(N+1));
disp('--- Hamming Window Coefficients ---');
disp(b_hamm);
%% --- Plot Frequency Responses ---
figure;
freqz(b_hamm,1);
title('Low-pass FIR using Hamming Window');

```

Output :



Expt 4 Design of Low Pass Filter using Windowing Techniques

Date: 4.3. Design and Analysis of an FIR Low-Pass Filter Using the Hanning Window Technique in MATLAB

Program :

```

clc;
clear;
close all;
%% --- Input Parameters ---
fp = 1000; % Passband cutoff (Hz)
fs = 4000; % Sampling frequency (Hz)
N = 50; % Filter order (even number preferred)

```

```

wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)
%% --- FIR Low Pass using Hanning Window ---
b_hann = fir1(N, wc, 'low', hanning(N+1));
disp('--- Hanning Window Coefficients ---');
disp(b_hann);
%% --- Plot Frequency Responses ---
figure;
freqz(b_hann,1);
title('Low-pass FIR using Hanning Window');

```

Output :

